Problem-Solution Fit

Date	02 JULY 2026
Team ID	CCAP-01
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

1. CUSTOMER SEGMENT(S) [CS] Bank credit analysts reviewing applications; financial compliance officers screening high-risk applicants; prospective customers applying for a new credit card.	**5. CUSTOMER LIMITATIONS EG. BUDGET, DEVICES [CL]** Banks have limited staff/time to manually review thousands of daily applications; applicants may lack complete credit history, documentation, or awareness of eligibility criteria.	**9.CHANNELS OF BEHAVIOR** **OFFLINE** In-branch consultation where an analyst manually enters applicant details into the system on the customer's behalf.
2. PROBLEMS / PAINS + ITS FREQUENCY [PR] High daily volume of applications causes slow turnaround; manual review is inconsistent and prone to human error; occurs continuously, every business day, at scale.	**6. PROBLEM ROOT / CAUSE [RC]** No automated mechanism exists to combine financial and demographic inputs into a fast, consistent, data-driven decision, so the process cannot scale with application volume.	**10. AVAILABLE SOLUTIONS PROS & CONS [AS]** Manual underwriting: accurate but slow and cause errors Rule-based scoring engines: fast but rigid, can't adapt to complex patterns. Generic credit bureau scores: not tailored to bank-specific risk factors.
3. TRIGGERS TO ACT [TR] Surge in application backlog, rising rejection complaints, and compliance/audit pressure push the bank to seek an automated, scalable screening solution.	**7. BEHAVIOR + ITS INTENSITY [BE]** Analysts manually cross-check each applicant's income, loan balances, and credit inquiries against policy guidelines — a high-effort task repeated for every single application.	**11. YOUR SOLUTION [SL]** An ML-based credit approval system trained on historical applicant data using Logistic Regression, Random Forest, XGBoost, and Decision Tree. The best-performing model is deployed via a Flask web app and an IBM Watson Machine Learning pipeline, delivering instant, real-time approval/rejection predictions through an intuitive UI.
4. EMOTIONS BEFORE / AFTER [EM] Before: frustration and uncertainty from slow, opaque manual review. After: confidence and trust from fast, transparent, data-driven predictions.	**.8. CHANNELS OF BEHAVIOR [CH]** **ONLINE** Web application used by analysts and customers to submit applicant data and instantly receive predictions; cloud-hosted via IBM Watson ML for scalability.	